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M 208 CIRCULATING WATER PIPING

1 PROCESS DESCRIPTION

The steam turbine condenser is once through seawater cooled. Cooling water will be taken from the existing water intake and returned to the existing seawater discharge structures. The circulating water piping shall be designed for each Combined Cycle Block.

The steam turbine auxiliary coolers are cooled by a closed cooling water circuit (Demineralized water). The closed loop cooling water is in turn cooled by seawater.

The seawater received from the intake structure shall be passed through the screening trains for removing of all kinds of debris.

The cleaned seawater shall be passed to the cooling water pumps suction chambers and then pumped through the C.W. Pumps to the steam turbine condenser and to the auxiliary coolers, then discharged to the outfall channel.

GRP/ (FRP) pipes shall be used for the circulating water supply to the condenser, the discharge channel will be a concrete channel.

As an alternative, carbon steel pipes shall be used for circulating water system with proper surface coating (3 Layer Polyethylene) for outside and rubber lining for inside Circulating Water piping and shall be completed with cathodic protection system.

The seawater shall be treated by hypochlorite injection to control organic substances and the growth of mussels, barnacles etc.

2 SCOPE OF WORK

The Bidder shall supply the above described circulating water piping system in accordance with part 4, subsection 1.2 “Summary of Work”.

The related civil works and structures of intake system, screening and pumping station and outfall system are specified in Sections M-206, M-207, S-001

The scope of supply shall include, but not be limited to the following:

- Piping from the CW pump discharge nozzles, to the main condenser inlet water boxes and from main condenser outlet water boxes to the circulating water discharge channel, and piping branches from the condenser inlet piping to the condenser outlet piping through the closed cooling water circuit heat exchangers and piping to the chlorination plant.
- Motor operated valves at CW pump discharges, and CW header main condenser inlets and outlets and CW inlets and outlets.
- Expansion joints at CW pump discharges, main condenser inlets and outlets, and where required by the piping design.
- Instrumentation and control devices
- Electrical equipment

3 CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5.”Codes and Standard”. The Codes and Standards of the following organizations extracted from subsection 2.5 and other international equivalent standards are Specifically applicable to the design, manufacture and testing of the equipment included in this Specification:

ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
AWWA	American Water Works Association
HIS	Hydraulic Institute Standards
IEC	International Electrotechnical Commission
NEMA	National Electrical Manufacturers

SSPC	Structural Steel Painting Council
DIN	Deutsches Institut Für Normung

The Bidder may use other standards not mentioned in this specification subject to be approved by the Owner. The Bidder shall provide those standards as applicable.

4 OPERATING AND DESIGN CONDITIONS

The circulating water main piping shall be designed for a maximum water velocity of 2.2 m/s. Maximum velocity all other C.W. piping shall be according to Part 4, specification M-001.

The design pressure range shall be the pressures corresponding to pump shut off head and full vacuum. In addition, peak pressures and vacuum due to valve and pump water hammer transient effects shall be considered in establishing the maximum design pressure.

The sea water analysis is specified in Part 4, Subsection 1.

5 DETAILED REQUIREMENTS

5.1 General

The following sections describe the technical requirements for the equipment and facilities to be provided by the Bidder for the circulating water (seawater) piping systems. The works to be provided shall be complete in every respect for the safe and faultless operation and easy maintenance of the complete systems and equipment, even if not expressly specified.

The requirements specified in M-001 „Piping“ shall apply and, where applicable, further regulations from the other sections of the Specification.

Piping to be installed in the pump house, and at the connections of equipment in the steam turbine building shall be above ground.

The other sections of piping shall be buried underground.

5.2 Piping

Machine made reinforced thermosetting resin pressure pipes and fittings adequate for the design conditions shall be used for the above ground pipework systems and underground pipework systems of the circulating water piping.

The material, subject of this specification, will be fabricated and tested in accordance with the latest edition, revision or addendum of the referenced codes, specifications or standards. They form a part of this specification.

- AWWA Standards:

AWWA C950-88	Standard for glass fibre reinforced thermosetting resin pressure pipe.
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- ASTM Standards:

B.31.3	Chapter VII
E203	Moisture content of resin
D543	Test for resistance of plastics to chemical reagents
D581	Testing of resins
D638	Test for tensile properties of plastics
D695	Test for compressive properties of rigid plastics
D833	Definitions of terms relating to plastics
D1544	Gardner color comparison of resin
D1598	Test for time-to-failure of plastic pipe under long term hydrostatic pressure
D1599	Test for short-time rupture strength of plastic pipe, tubing and fittings.
D1600	Abbreviations of terms relating to plastics
D1639	Acid number of resin
D1898	Recommended practice for sampling of plastics
D2105	Test for longitudinal tensile properties of reinforced, thermosetting plastic pipe

D2143	Test for cyclic pressure strength of reinforced, thermosetting plastic pipe
D2290	Test for apparent tensile strength of ring or tubular plastics by split disc method
D2310	Classification of machine made reinforced thermosetting resin pipe.
D2343	Tensile properties of glass fibre
D2393 & 445	Viscosity of resin
D2517	Specification for reinforced thermosetting plastic gas pressure piping and fittings
D2563	Classification of visual defects in glass reinforced laminates and pieces built from same.
D2583	Standard test method for indentation hardness of rigid plastics by means of a Barcol impresser
D2584	Standard method of test for ignition Loss of cured reinforced resins
D2772	Test for sieve analysis of electrical grade magnesium oxide
D2924	External pressure resistance
D2992	Obtaining hydrostatic design basis for reinforced thermosetting resin pipe and fittings
D2995	Standard specification for filament wound reinforced thermosetting resin pipes
D3171	Fiber content of compound reinforced resin materials
D3567	Standard method of determining dimension of reinforced thermosetting resin pipe and fittings
D3839	Standard recommended practice for underground installation of flexible reinforced thermosetting pipe and reinforced plastic mortar pipe.
NSF	Standard No.14 - plastic piping system components and related materials.

- British Standards

BS3396	Woven glass fibre fabrics for plastic reinforcement
BS3496	Glass fibre chopped, strand mat for the reinforcement of polyester resin systems
BS3691	Glass fiber roving for the reinforcement of polyester resin systems
BS4504	Flanges
BS5480	GRP - pipes and fittings for use of water supply or sewerage
BS6031	1981 - code of practice for earth works.

The pipe manufacturer shall design all GRP pipe, fittings, and joints to resist loads imposed on the piping system in accordance with the referenced engineering drawings and in keeping with the design criteria described herein.

The design and erection of the GRP piping systems shall be in accordance with the requirements of the AWWA C950-81 standard.

Cemented joints in GRP pipework is not acceptable. Keylock or laminated joints are required.

The final design must reflect consideration of both the combined longitudinal loads as well as the combined circumferential loads which will be imposed on the installed GRP system.

All design analysis calculations by the manufacturer shall be submitted to the Owner for evaluation and approval.

The design for above ground pipes shall consider pigments for UV protection and internal algae growth prevention.

5.2.1 Fabrication Methods

5.2.1.1 Pipes

Pipes subject of this specification shall be manufactured by the continuous filament winding process (Drosthlo) or by the helical filament winding process.

In both processes the continuous fibrous glass strand rovings are wound on to the outside of a mandrel in a predetermined pattern under controlled tension.

The rovings are or helical wound with a winding angle depending on the size of the pipe and/or the application.

The inside diameter of the finished pipe is fixed by the mandrel outside diameter.

The outside diameter of the finished pipe is determined by the amount of material that is wound on the mandrel.

Pipes shall meet ASTM D2310 classification code and shall be classified with a designation code in accordance with ASTM D2996.

5.2.1.2 Fittings

Fittings subject to this specification are 11°, 15°, 22°, 30°, 45° and 90° elbows, concentric reducer, tees, flanges, adapters and couplers.

Fittings in the diameter range up to 300mm are filament wound, using preimpregnated roving tape (woven cloth), which will be applied on to the outside of a mandrel in a predetermined pattern under controlled tension. Elbows in the diameter range from 100 mm up to 300mm can also be manufactured by filament winding of continuous fibrous glass strand roving. In the diameter range up to 300mm the following fittings shall be standard available: 45° and 90° elbows, concentric reducers, 45° laterals, tees flanges, adapters and couplers.

For the fittings up to 300 mm the inside diameter is fixed by the mandrel outside diameter.

The outside diameter of the fitting is determined by the amount of material that is wound on the mandrel.

Fittings in the diameters ranging from 350 mm up to 2000 mm are manufactured from pipe parts which are fitted together first and then wrapped round with preimpregnated woven cloth.

Pipe parts will meet the pipe requirements.

In this diameter range the following fittings shall be standard available: 11°, 15°, 22°, 30°, 45° and 90° elbows, concentric reducers, tees and flanges filament wound GRP-pipe - wall structure.

Resin-rich inner liner (corrosion barrier):

- The interior wall of the filament wound pipe (sometimes called the liner or internal corrosion barrier having an approximately thickness of 1.5 to 2.0 mm) shall be constructed so as to have a smooth, resin-rich surface. The innermost 0.5 to 0.8 mm surface shall be reinforced with one or two highly corrosion resistant veils of “C” glass. This first surfacing layer of inner liner shall contain approximately 90% of resin material.
- The remainder of the liner 1.0 to 1.2 mm shall consist of 75% of resin and may incorporate two layers of 450 g/m² powder bound “E” glass chopped strand mat having a glass content between 25 and 35% by weight on the equivalent of chopped strand “E” glass depending on the method of manufacture.
- The liner thickness for epoxy pipes shall have an approximate thickness of 0.5 mm. This layer shall be reinforced with one or two highly corrosion resistant veil of “C” glass.

5.2.2 Thermal and Other physical properties

The Bidder shall complete the following :

Coefficient of higher thermal expansion	mm/m.K
Thermal conductivity	W/m.K
Specific heat	J/kg.K
Glass ratio (by weight)	%
Specific gravity	
Maximum support spacing (above ground)	m
Maximum allowable deflection of pipe (full) and under pressure	mm
Weight per meter (pipe full)	kg
Weight per meter (empty)	g

5.2.3 Design Procedure

The design procedure for GRP/FRP systems involves establishment of design conditions, selection of pipe class and corresponding pipe properties, selection of installation parameters, and performance of pertinent calculations to assure that the design requirements of Sec. A2 of ANSI/AWWA C950-88 are satisfied or ANSI B.31.3 whichever is applicable.

If the results of any calculation indicate that a requirement is not satisfied, it may be necessary to upgrade installation parameters or select pipe with different properties, or both, and revise pertinent calculations.

5.2.4 Design Conditions

The Bidder shall design the pipework in accordance with the prevalent conditions.

5.2.4.1 Head losses

Hydraulic head loss due to pipe friction shall be calculated with a roughness coefficient representing adverse operational conditions, considering marine growth, if applicable. The roughness coefficient shall be approved by the Owner.

5.2.4.2 Surge pressures

Excessive surge pressures will be investigated in the detail design phase and the condition eliminated or automatic surge- pressure relief provided, so that the maximum working pressure will not be exceeded.

Resistance to erosion due to suspended solids will be investigated and reported by the manufacturer.

5.2.4.3 Joints

Details of joints and fully dimensioned drawings of joints. These drawings shall show the main pipe barrel and the complete joint between two pipe ends. Dimensions of internal diameters, wall thickness, length of joint and the extent of all shaping, cut outs, etc. should be indicated on the drawing together with the manufacturing tolerances on these dimensions.

5.2.4.4 Documents to be issued

- Details of all fittings including fully dimensioned drawings.
- Details of all standard pipe supports.
- Weights of all pipes and fittings.
- Calculations to prove the adequacy and to show the acceptable initial and long term deflections of each class of each diameter of installed pipe.
- A summary of the calculations for each class of each diameter of pipe.
- Results of all type tests as required in inspection and testing.
- Surge analysis for above ground pipes.

Should any details of the pipes and fitting be altered in any way during manufacture from those proposed during the mobilization period and approved by the Owner then the Bidder shall submit for the Owner's approval the revised details and test results of the pipe wall thickness and for stiffness where altered. Calculations for the modified pipes and fittings shall also be submitted.

All shop drawings are to be submitted to the Owner for review and approval prior to start of work.

No pipes or fittings will be accepted on site until the data appropriate to pipes of that class and diameter has been submitted to and approved by the Owner.

5.2.4.5 Handling and Storage

All details relating to the handling and storage of GRP/FRP-pipes and fittings including:

- Pipe storage facilities
- Site transport
- Lifting methods

Full details with sketches of installation procedures and such other methods of pipeline construction proposed.

Testing procedures and details, with sketches, of the deflection measurement apparatus.

Procedures for erecting the pipes into concrete structures, together with sketches showing site fit-ups etc.

Details of in-situ joint construction involving hand lay up butt and wrapped joints.

Trenching and compaction of backfill procedures where applicable.

Reasonable care must be exercised to prevent damage to the pipe during the many operations prior to and during shipment and at the installation site. Acceptable practices must be followed at the manufacturer's plant (handling, storage and loading for shipment) as well as at the installation site (unloading, storage, stringing and installation) to prevent damage from point loading and impact.

Each piece of pipe and fitting shall be marked at least once with the following information in such a manner that it remains legible and normal handling and installation practices:

- nominal pipe size
- nominal pipe length
- pressure class
- manufacturer's name (or trade mark)

Specially written manufacturer's instructions are to be submitted for approval and followed in all cases.

5.2.4.6 Hydraulic test

100% of the pipes shall be subjected to an internal hydraulic pressure test at the manufacturer's works prior to delivery. The test shall be applied to a pressure equal to 1.5 times design pressure stated for each classification of pipe. The test pressure shall be applied for a minimum period of 10 minutes without signs of leakage.

In addition to the above the first pipe and every thirtieth thereafter of each class and diameter shall be maintained at test pressure for a minimum of 4 hours without signs of leakage.

5.3 Valves

The valves shall be designed, manufactured, supplied and tested in accordance with the requirements of Part 4 valves specification M-004.

5.3.1 Materials

Materials for the valve body, disc, shaft and seat shall be suitable for seawater services and shall conform to the latest edition of the specified ASTM material specification or equivalent in all regards. Material Test Reports shall be submitted for the body, disc, and shaft for all valves. Material shall be the most suitable for the service conditions. The selected materials shall be of high resistance to sea water corrosion and subject to Owner's approval.

Valve seats shall be leak-tight at the design pressure and temperatures specified.

The transient surge pressure shall not affect the valve operability or reliability after the transient pressure occurs.

Valve bearing shall be designed the most suitable for service condition.

5.3.2 Valve Operation And Maintenance

Valves and operators shall be so constructed that valve disc will remain in any set position, full open to full close. Valves may at times be used to throttle the flow. Bidder shall identify all valves subject to throttling for long deviations, e.q. following pump trips, and shall design and certify the valves for throttling service.

Valves shall be of a design which permits adjustment, removal and replacement of the seats, discs, shafts, bearings, and seals in the field.

Each valve shall be equipped with mechanical position indicators. Valve operator shall be equipped with hand wheels for manual operation and shall have provision to be driven by a hand held pneumatic drive.

Bidder shall be fully responsible for each of the following requirements for each motor operated valve:

Factory set the motor operator torque switch to properly control and limit the opening and closing travel of the valve under the specified operating conditions.

Factory set the motor operator limit switches, including intermediate position as shown on Bidder's logic diagrams.

Designate the torque switch spring number and document the direct output torque reading in Newton-meters for each setting.

Certify that the valve motor operator is properly sized and properly adjusted to assure that the valve will seat absolutely tight on each closure.

Such special tools as may be necessary to adjust or dismantle the equipment shall be furnished by Bidder.

5.4 Expansion Joints

Materials shall be suitable for the specified operating conditions.

The maximum transient surge pressure shall not damage the expansion joints.

Bidder shall submit a detailed description of the procedure required for attachment of all expansion joint assemblies to the piping.

Expansion joints shall be of design, type and face-to-face dimension as approved by the Owner.

Expansion joints shall be flanged on both ends.

6 PERFORMANCE TESTING AND GUARANTEES

Refer also to Part 4 Subsection 2.6 „Performance Testing, Guarantees and first Inspection“ and section 3, M-014 „Inspection and testing“

Piping, valves, expansion joints, fittings and accessories shall be tested at shop and site in accordance with approved standards.

Inspection Test Plan and Inspection Test Procedure to be applied at site and shop shall be submitted to the Owner for approval.

All other C.W. piping shall be subject to inspection and testing requirements in Part 4, Specification M-001.

7 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality control shall conform to the requirements of Part 4, Subsection 2.7.

The quality control procedures and quality control documentation as specified shall be provided by the Bidder including all other documentation required by applicable codes and standards.

8 INSTRUMENTATION AND CONTROL

For detail Instrumentation and Control requirements, see Part 4, I & C Technical Specifications.

9 ELECTRICAL SPECIFICATIONS

For detail Electrical requirements, see Part 4, Electrical Technical Specification.

Valve electric motors shall per E-210 Low Voltage Motors.

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